

IIT Jammu Summer School 2026

Course Name: IoT and Drone Building

Fundamentals of Drones and UAV Systems

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Q1. What is a drone (UAV)? Explain its definition, history, and evolution.

A **Drone** or **Unmanned Aerial Vehicle (UAV)** is a powered aircraft that operates without a human pilot, crew, or passengers on board. Instead of being flown by a person inside the

aircraft, a drone is either controlled remotely by an operator or operates autonomously using onboard systems. A UAV is one part of a larger **Unmanned Aircraft System (UAS)** or **Remotely Piloted Aircraft System (RPAS)**, which also includes the ground control station, communication links, and other supporting equipment required for safe operation.

History of UAVs

The development of drones has progressed through several stages, transforming them from experimental military devices into versatile tools used across numerous industries.

1. Early Foundations (1800s – Early 1900s):

The earliest known military use of an unmanned flying device occurred in **1849**, when Austrian forces launched incendiary balloons against Venice. Later, in **1903**, Spanish engineer **Leonardo Torres Quevedo** developed the **Telekino**, a radio-controlled system designed to remotely operate airships, laying the foundation for modern remote-control technology.

2. World Wars and Target Drones (1910s – 1940s):

During **World War I**, research focused on developing unmanned aircraft for military training and combat. Examples include **A. M. Low's Aerial Target** and the **Kettering Bug**, an early unmanned aircraft designed to deliver explosives. In **1935**, the British **de Havilland Queen Bee** entered service as a pilotless target aircraft, and it is widely believed that the term "**drone**" originated from its name. During **World War II**, unmanned technology advanced further with systems such as Germany's **V-1 flying bomb**.

3. Tactical and Reconnaissance Era (1950s – 1980s):

After the **U-2 spy plane incident** in 1960, the United States expanded its UAV programs, leading to reconnaissance aircraft such as the **Ryan Model 147**, which was widely used during the Vietnam War. In the 1970s, **Israel** introduced tactical UAVs equipped with cameras capable of providing real-time battlefield surveillance. By **1975**, the **Tadiran Mastiff** became one of the first modern battlefield drones, featuring extended flight endurance and live video transmission.

4. Commercialization and Modern Era (2000s – Present):

Advances in microcontrollers, sensors, batteries, and wireless communication significantly reduced the cost of drone technology, making it accessible for civilian use. A major milestone was the launch of the **DJI Phantom** in **2013**, which popularized consumer drones for aerial photography, videography, and recreational flying.

Modern Evolution of Drones

Today, drones have evolved from simple remotely controlled aircraft into intelligent autonomous systems. Modern UAVs incorporate technologies such as **Artificial Intelligence (AI)**, **computer vision**, and **machine learning**, enabling them to detect obstacles, analyze their surroundings, and perform complex tasks with minimal human intervention. As a result, drones are widely used in sectors such as agriculture, logistics, infrastructure inspection, disaster management, surveillance, search and rescue, and environmental monitoring. They have also expanded into space exploration, with **NASA's Ingenuity helicopter** demonstrating powered flight on Mars. Today, drones are increasingly being integrated into global airspace systems, becoming an essential part of modern transportation, industry, and public services.

Q2. List at least five real-world applications of drones.

Drones have become an important part of many industries due to their ability to perform tasks quickly, safely, and efficiently. Some of the major real-world applications of drones are:

1. Precision Agriculture

Drones are widely used in modern agriculture for monitoring crop health, precision spraying of pesticides and fertilizers, seed planting, and field mapping. They help farmers identify crop diseases early, optimize resource usage, and improve overall agricultural productivity.

2. Logistics and Delivery

Drones are increasingly being used for last-mile delivery of packages, food, and medical supplies. Companies such as Zipline use drones to deliver blood, vaccines, and emergency medicines to remote and hard-to-reach areas, reducing delivery time and improving healthcare accessibility.

3. Infrastructure Inspection and Maintenance

Drones equipped with cameras, thermal sensors, and LiDAR systems are used to inspect bridges, power lines, wind turbines, oil and gas pipelines, and other critical infrastructure. They reduce the need for dangerous manual inspections while improving efficiency and accuracy.

4. Public Safety and Disaster Management

During natural disasters and emergency situations, drones provide real-time aerial surveillance to assist in search and rescue operations, damage assessment, wildfire monitoring, flood management, and traffic surveillance. Their ability to access hazardous areas makes them valuable tools for emergency response teams.

5. Military and Defense

The defense sector extensively uses drones for intelligence, surveillance, reconnaissance (ISR), border monitoring, target tracking, and tactical missions. They can perform operations in high-risk environments without putting human lives in danger.

Other Applications

Apart from the above, drones are also used in aerial photography and filmmaking, environmental monitoring, scientific research, wildlife conservation, land surveying, drone racing, and entertainment through synchronized drone light shows.

Q3. Differentiate between: (a) UAV and Drone, (b) Fixed-wing and Multirotor drones, (c) Quadcopter and Hexacopter.

(a) UAV vs Drone

UAV (Unmanned Aerial Vehicle)

Drone

A technical term referring specifically to the aircraft that flies without a human pilot onboard.

A commonly used general term for an unmanned flying vehicle.

It represents only the flying vehicle.

It is an informal term often used by the public and media for UAVs.

Used mainly in engineering, aviation, and regulatory documents.

Used in everyday communication and commercial applications.

A UAV is a part of a larger Unmanned Aircraft System (UAS), which includes the ground control station, communication links, and other support equipment.

The term "drone" is often used to refer to the complete system, although technically it usually means the aircraft itself.

(b) Fixed-Wing Drone vs Multirotor Drone

Fixed-Wing Drone

Multirotor Drone

Generates lift using fixed wings similar to an airplane.

Generates lift using multiple rotating propellers.

Has long flight endurance and can cover large distances.

Has shorter flight time, typically around 15–30 minutes.

Cannot hover in one position.

Can hover and perform Vertical Take-Off and Landing (VTOL).

Requires a runway, catapult, or launcher for take-off and landing.

Can take off and land vertically without a runway.

Suitable for mapping, surveying, and long-range missions.

Ideal for aerial photography, inspection, surveillance, and confined-space operations.

(c) Quadcopter vs Hexacopter

Quadcopter

Has **four** rotors.

Simpler design and lower cost.

Carries relatively lighter payloads.

Less fault tolerant; failure of one motor usually results in loss of control.

Commonly used for recreational flying, photography, and small commercial applications.

Hexacopter

Has **six** rotors.

More complex and expensive.

Can carry heavier payloads such as professional cameras and sensors.

More reliable because it can often continue flying safely even if one motor fails.

Commonly used for industrial inspection, surveying, filmmaking, and professional aerial operations.

Q4. Explain how a quadcopter flies. Include lift, thrust, gravity, drag, propeller rotation, and clockwise (CW) / counter-clockwise (CCW) motors. Draw a neat diagram.

A quadcopter flies by balancing four fundamental forces of flight—**lift, thrust, gravity, and drag**—while controlling the speed of its four motors. The flight controller continuously adjusts the motor speeds to maintain stability and perform different flight movements.

1. Lift

Lift is the upward aerodynamic force generated by the rotating propellers. When the lift produced by the propellers is greater than the drone's weight, the quadcopter rises. If lift equals the weight, the drone hovers, and if lift becomes less than the weight, the drone descends.

2. Thrust

Thrust is the force produced by the motors and propellers that pushes air downward, allowing the drone to move through the air. By changing the thrust generated by individual motors, the quadcopter can move in different directions.

3. Gravity (Weight)

Gravity is the downward force acting on the drone due to Earth's gravitational pull. During stable hovering, the upward lift is equal to the downward force of gravity, keeping the drone in equilibrium.

4. Drag

Drag is the air resistance encountered by the drone while flying. As the drone moves faster, drag increases. The motors must generate additional thrust to overcome this resistance and maintain the desired speed.

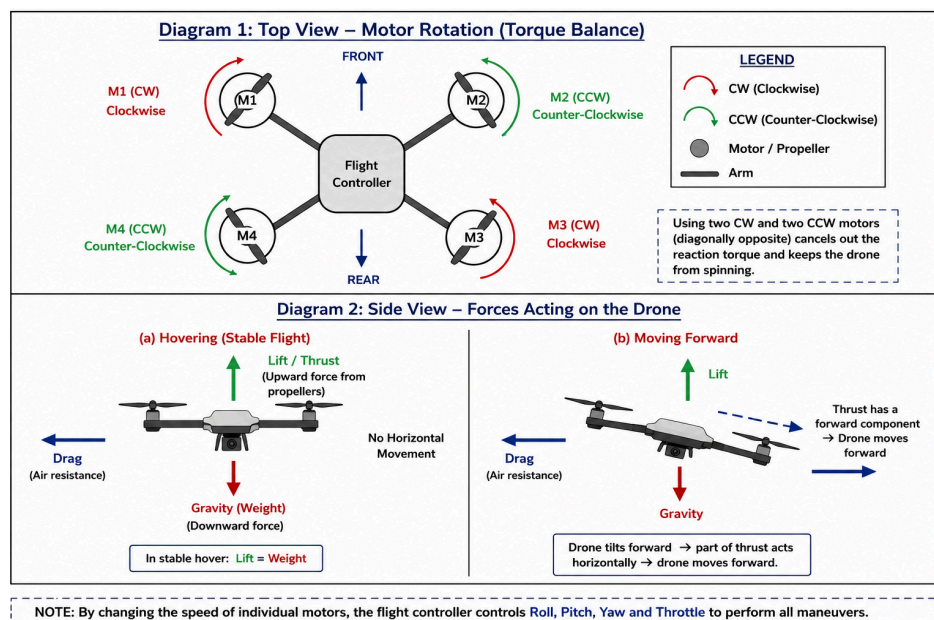
Propeller Rotation

A quadcopter has **four brushless motors**, each driving a propeller. To prevent the drone from spinning uncontrollably due to motor torque, two propellers rotate **Clockwise (CW)** while the other two rotate **Counter-Clockwise (CCW)**. The clockwise and counter-clockwise motors are placed diagonally opposite each other, allowing their torques to cancel out and keep the drone stable.

Drone Movement

The flight controller continuously adjusts the speed of individual motors to control the drone's movement.

- **Take-off:** All four motors increase speed equally, generating lift greater than the drone's weight.
- **Hovering:** All four motors rotate at equal speed, producing lift equal to gravity.
- **Yaw (Rotation):** The flight controller increases the speed of one pair of motors (CW or CCW) while decreasing the speed of the opposite pair, creating a torque imbalance that rotates the drone about its vertical axis.
- **Pitch and Roll:** To move forward, backward, left, or right, the motors on one side spin faster while the motors on the opposite side spin slower. This tilts the drone and directs part of the thrust horizontally, causing movement in the desired direction.



Q5. Explain how a drone performs take-off, landing, hovering, move forward, move backward, move left, move right, and rotate (yaw). Use suitable diagrams.

A quadcopter performs different flight maneuvers by varying the speed of its four brushless motors using **Electronic Speed Controllers (ESCs)**. The flight controller continuously processes data from sensors such as the **IMU**, **gyroscope**, **accelerometer**, and **GPS** and adjusts the motor speeds to maintain stability and achieve the desired movement.

1. Take-off

During take-off, all four motors increase their speed simultaneously, producing greater thrust and lift. When the upward lift becomes greater than the drone's weight, the quadcopter rises vertically into the air.

2. Hovering

To hover, all four motors rotate at equal speeds so that the upward lift is exactly equal to the downward gravitational force. The flight controller continuously makes small adjustments to maintain a stable position and altitude despite external disturbances such as wind.

3. Landing

For landing, the flight controller gradually reduces the speed of all four motors. As the lift becomes slightly less than the drone's weight, the quadcopter descends smoothly until it reaches the ground.

4. Moving Forward (Pitch)

To move forward, the rear motors rotate faster while the front motors rotate slower. This causes the drone to tilt forward (pitch), directing part of the thrust horizontally and moving the drone forward.

5. Moving Backward

To move backward, the front motors rotate faster while the rear motors rotate slower. The drone tilts backward, causing a horizontal thrust component that moves it in the backward direction.

6. Moving Left (Roll)

To move left, the motors on the right side rotate faster while those on the left side rotate slower. The drone rolls toward the left, allowing part of the thrust to move it sideways.

7. Moving Right (Roll)

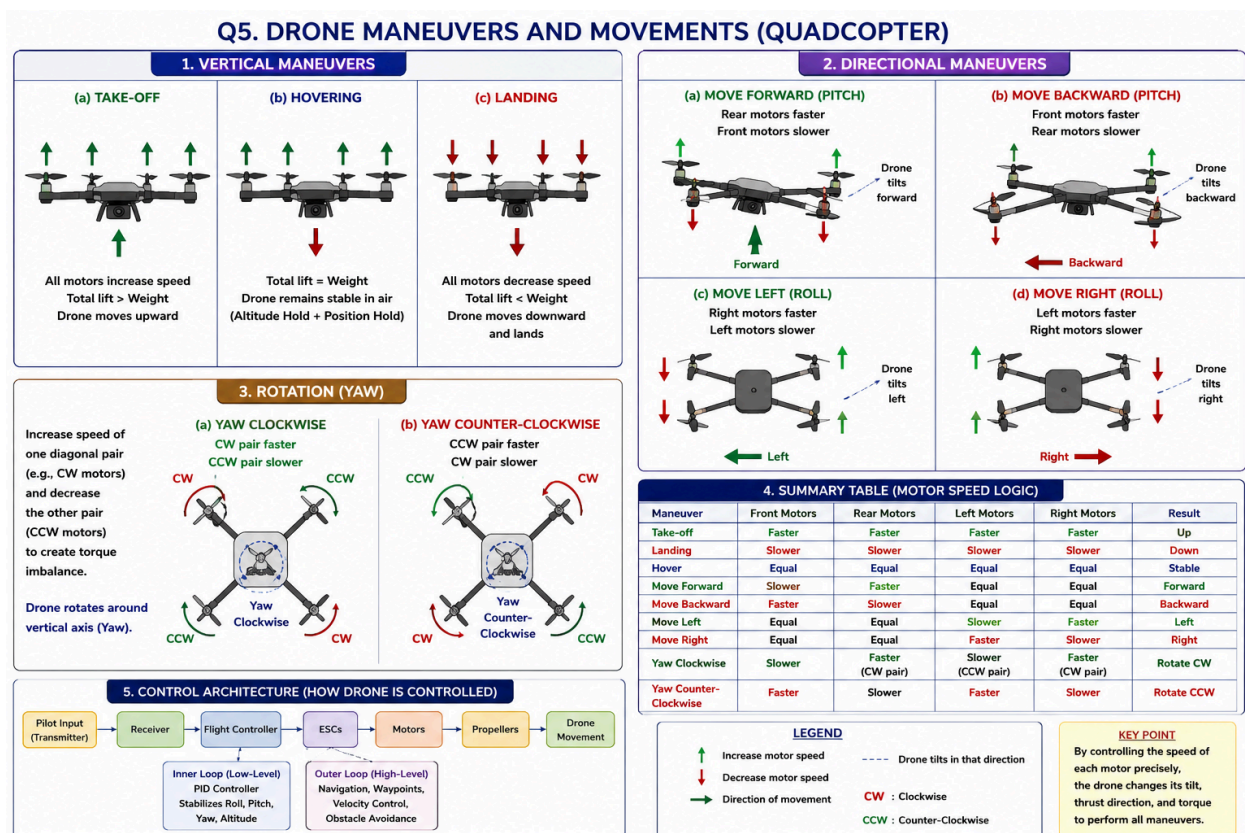
To move right, the left-side motors rotate faster while the right-side motors rotate slower. This produces a roll toward the right, resulting in rightward movement.

8. Rotation (Yaw)

Yaw is the rotational movement of the drone about its vertical axis. A quadcopter has two **Clockwise (CW)** and two **Counter-Clockwise (CCW)** rotating propellers. To rotate, the flight controller increases the speed of one pair of motors while decreasing the speed of the opposite pair. This creates a torque imbalance, causing the drone to rotate without significantly changing its altitude.

Summary

The flight controller coordinates all these movements by continuously adjusting the speed of individual motors through the ESCs. By controlling the balance of lift, thrust, gravity, and torque, the drone can perform stable take-off, landing, hovering, directional movement, and rotation with high precision.



Q6. Draw a labeled block diagram of a drone and explain the function of each component: flight controller, ESC, brushless motors, propellers, battery, power distribution board, GPS module, radio receiver, frame, telemetry module.

A drone consists of several electronic and mechanical components that work together to achieve stable flight, navigation, and communication. The block diagram illustrates how these components are interconnected, while their functions are explained below.

1. Flight Controller (FC)

The **Flight Controller (FC)** is the brain of the drone. It receives data from onboard sensors such as the IMU, GPS, and radio receiver, processes this information, and sends control signals to the Electronic Speed Controllers (ESCs). It is responsible for maintaining stability, controlling roll, pitch, yaw, altitude, and executing autonomous flight operations.

2. Electronic Speed Controller (ESC)

The **Electronic Speed Controller (ESC)** regulates the speed of each brushless motor. It receives Pulse Width Modulation (PWM) signals from the flight controller and adjusts the motor's RPM accordingly. By varying motor speeds, the ESC enables the drone to hover, turn, climb, descend, and perform other maneuvers.

3. Brushless Motors (BLDC Motors)

Brushless DC (BLDC) motors convert electrical energy from the battery into mechanical rotation. They drive the propellers and generate the thrust required for flight. BLDC motors are preferred because they are lightweight, energy-efficient, reliable, and require less maintenance than brushed motors.

4. Propellers

Propellers are attached to the brushless motors and generate lift and thrust by pushing air downward. The rotation of the propellers produces the aerodynamic forces needed for take-off, hovering, movement, and landing.

5. Battery

The drone is powered by a **Lithium-Polymer (Li-Po)** battery, which supplies electrical energy to all onboard components. Battery capacity directly affects the drone's flight time, payload capability, and overall performance.

6. Power Distribution Board (PDB)

The **Power Distribution Board (PDB)** distributes electrical power from the battery to the flight controller, ESCs, GPS module, telemetry module, and other electronic components. It acts as the central power hub of the drone.

7. GPS Module

The **Global Positioning System (GPS) Module** provides real-time location information, including latitude, longitude, altitude, and speed. It enables waypoint navigation, autonomous flight, Return-to-Home (RTH), and accurate position holding.

8. Radio Receiver

The **Radio Receiver** receives wireless control signals from the pilot's transmitter and forwards these commands to the flight controller. This allows the pilot to manually control the drone during flight.

9. Frame

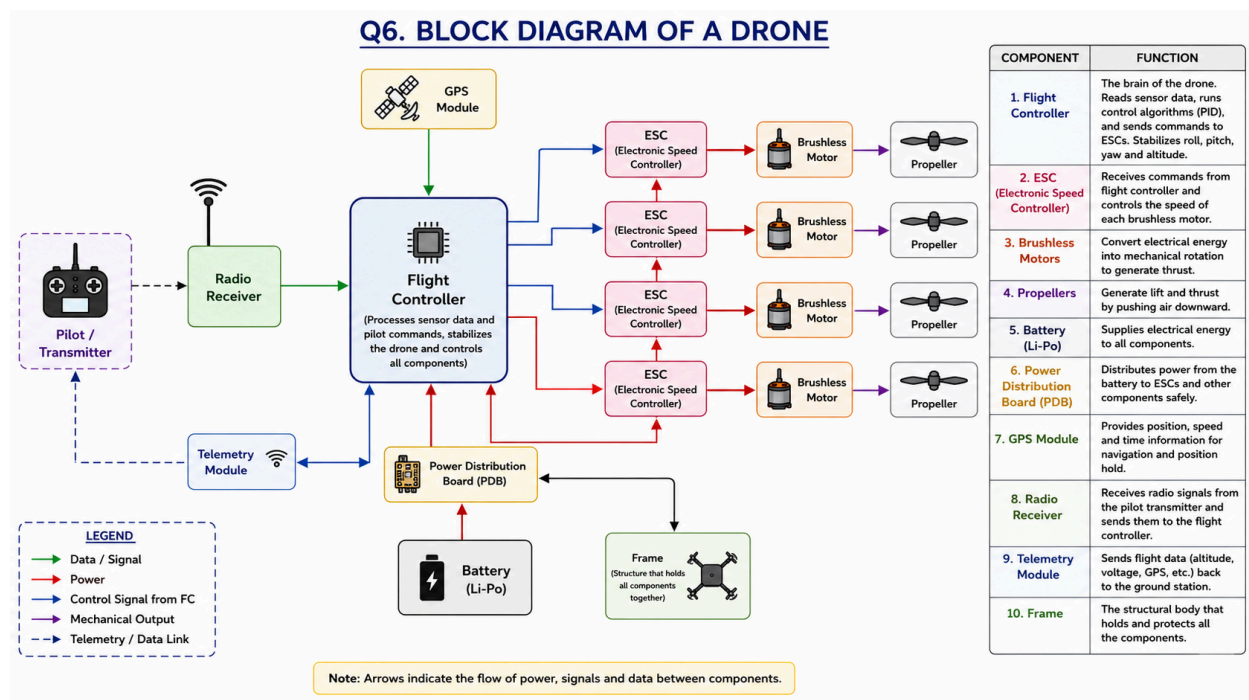
The **Frame** forms the structural body of the drone. It supports and protects all components, including the motors, battery, flight controller, and sensors, while maintaining a balance between strength and lightweight construction.

10. Telemetry Module

The **Telemetry Module** provides two-way communication between the drone and the ground control station. It continuously transmits important flight information such as battery voltage, altitude, speed, GPS coordinates, and system status, allowing the operator to monitor the drone in real time.

Working of the Block Diagram

The pilot sends commands through the **radio transmitter**, which are received by the **radio receiver**. These commands, along with data from the **GPS module** and onboard sensors, are processed by the **flight controller**. The flight controller then sends PWM control signals to the **ESCs**, which regulate the speed of the **brushless motors**. The motors rotate the **propellers**, producing the lift and thrust required for flight. Electrical power is supplied by the **Li-Po battery** through the **Power Distribution Board (PDB)**, while the **telemetry module** continuously sends flight information back to the ground station. All these components are securely mounted on the **frame**, which provides structural support to the drone.



Q7. Explain the role of the following components: flight controller, ESC, BLDC motor, propeller, Li-Po battery, GPS, radio transmitter, frame, battery eliminator circuit (BEC), telemetry module.

The efficient operation of a drone depends on the coordinated functioning of several electronic and mechanical components. The role of each component is explained below.

1. Flight Controller (FC)

The **Flight Controller (FC)** is the brain of the drone. It processes data from sensors such as the accelerometer, gyroscope, barometer, and GPS to stabilize the aircraft and control its roll, pitch, yaw, and altitude. It also executes autonomous flight functions such as waypoint navigation and Return-to-Home (RTH).

2. Electronic Speed Controller (ESC)

The **Electronic Speed Controller (ESC)** regulates the speed of each brushless motor by receiving PWM control signals from the flight controller. It adjusts the motor RPM to enable smooth take-off, hovering, landing, and directional movement.

3. Brushless DC (BLDC) Motor

A **Brushless DC (BLDC) Motor** converts electrical energy into mechanical rotation to drive the propellers. BLDC motors are preferred because they are lightweight, efficient, reliable, produce less noise, and require minimal maintenance.

4. Propeller

The **Propeller** generates lift and thrust by rotating at high speed and pushing air downward. By changing the speed of individual propellers, the drone can climb, descend, hover, and move in different directions.

5. Lithium-Polymer (Li-Po) Battery

The **Li-Po Battery** is the primary power source of the drone. It supplies electrical energy to all onboard components, including the motors, flight controller, ESCs, GPS, and communication modules. Its high energy density provides longer flight time while maintaining a lightweight design.

6. Global Positioning System (GPS)

The **GPS Module** provides real-time position information, including latitude, longitude, altitude, and speed. It enables autonomous navigation, waypoint missions, Return-to-Home (RTH), and accurate position holding during flight.

7. Radio Transmitter

The **Radio Transmitter** is the handheld controller operated by the pilot. It sends wireless control commands to the drone, allowing the pilot to control throttle, roll, pitch, and yaw during manual flight.

8. Frame

The **Frame** is the structural body of the drone that supports and protects all electronic and mechanical components. A strong yet lightweight frame improves stability, durability, and overall flight performance.

9. Battery Eliminator Circuit (BEC)

The **Battery Eliminator Circuit (BEC)** converts the high voltage supplied by the Li-Po battery into a lower, stable voltage required by the flight controller, receiver, GPS, and other electronic components. It eliminates the need for separate batteries for these devices and ensures a reliable power supply.

10. Telemetry Module

The **Telemetry Module** establishes a two-way communication link between the drone and the ground control station. It continuously transmits important flight information such as battery voltage, altitude, GPS coordinates, speed, and system status, enabling real-time monitoring and mission management.

Q8. Explain the working principle and application of IMU, accelerometer, gyroscope, magnetometer, barometer, GPS, optical flow sensor, ultrasonic sensor, LiDAR, and camera. For each sensor mention purpose, working principle, and application.

Sensor	Purpose	Working Principle	Application
Inertial Measurement Unit (IMU)	Provides attitude control, stabilization, and state estimation.	Combines a 3-axis accelerometer and a 3-axis gyroscope (and sometimes a magnetometer and barometer) to measure the drone's movement and orientation.	Used for stable flight, attitude control, and maintaining balance.
Accelerometer	Measures linear acceleration and determines the drone's orientation.	Detects acceleration along the X, Y, and Z axes by sensing the force of gravity and motion.	Used for self-leveling, tilt detection, and flight stabilization.
Gyroscope	Measures angular velocity (rate of rotation).	Detects rotational movement around the X, Y, and Z axes.	Used to stabilize roll, pitch, and yaw during flight.
Magnetometer	Determines the drone's heading relative to magnetic north.	Detects Earth's magnetic field and acts as a digital compass.	Used for yaw control, waypoint navigation, and maintaining flight direction.

Barometer	Measures altitude.	Calculates height by measuring changes in atmospheric pressure.	Used for altitude hold and smooth vertical flight.
GPS (Global Positioning System)	Provides real-time position information.	Receives signals from multiple satellites to calculate latitude, longitude, and altitude.	Used for waypoint navigation, Return-to-Home (RTH), geofencing, and autonomous flight.
Optical Flow Sensor	Detects movement relative to the ground.	Captures consecutive ground images and compares them to calculate horizontal motion.	Used for indoor flight, GPS-denied navigation, and low-altitude hovering.
Ultrasonic Sensor	Measures short-range distance to nearby objects or the ground.	Emits ultrasonic sound waves and measures the time taken for the echo to return.	Used for obstacle avoidance, precision landing, and maintaining ground clearance.
LiDAR (Light Detection and Ranging)	Creates accurate 3D maps and measures distance.	Emits laser pulses and calculates distance using the time taken for reflected light to return.	Used for SLAM, terrain mapping, infrastructure inspection, and precision agriculture.
Camera	Captures visual information for monitoring and imaging.	Uses image sensors (RGB, infrared, or multispectral) to convert light into digital images or videos.	Used for aerial photography, surveillance, crop monitoring, mapping, inspection, and target tracking.

Q9. Compare: (a) GPS vs Optical Flow, (b) LiDAR vs Ultrasonic, (c) Gyroscope vs Accelerometer.

(a) GPS vs Optical Flow Sensor

GPS

Provides global position and navigation information.

Works by receiving signals from GPS satellites.

Best suited for outdoor navigation and long-distance missions.

Used for waypoint navigation, Return-to-Home (RTH), and geofencing.

Optical Flow Sensor

Measures movement relative to the ground.

Works by analyzing consecutive images of the ground to detect motion.

Best suited for indoor flight and GPS-denied environments.

Used for stable hovering, position hold, and low-altitude flight.

(b) LiDAR vs Ultrasonic Sensor

LiDAR

Uses laser light to measure distance.

Provides high-accuracy 3D mapping and long-range sensing.

More accurate but comparatively expensive.

Used for SLAM, terrain mapping, and infrastructure inspection.

Ultrasonic Sensor

Uses ultrasonic sound waves to measure distance.

Provides short-range distance measurement.

Less accurate but inexpensive and lightweight.

Used for obstacle avoidance, precision landing, and ground distance measurement.

(c) Gyroscope vs Accelerometer

Gyroscope

Measures angular velocity (rotation).

Detects rotational movement around the X, Y, and Z axes.

Accelerometer

Measures linear acceleration and tilt.

Detects acceleration along the X, Y, and Z axes.

Helps stabilize roll, pitch, and yaw during flight.

Determines orientation relative to gravity and enables self-leveling.

Corrects unwanted rotations caused by wind or disturbances.

Assists in maintaining balance and estimating the drone's attitude.

Q10. Explain the concept of 6 Degrees of Freedom (6DOF). Include translation, rotation, X-axis, Y-axis, Z-axis. Draw suitable diagrams.

6 Degrees of Freedom (6DOF) refers to the six possible ways in which a drone can move and change its orientation in three-dimensional (3D) space. A drone achieves 6DOF using an **Inertial Measurement Unit (IMU)**, which typically consists of a **3-axis accelerometer** and a **3-axis gyroscope**. These sensors continuously measure the drone's movement and orientation, enabling stable flight and accurate navigation.

The six degrees of freedom are divided into **three translational movements** and **three rotational movements**.

1. Translational Movements (Linear Motion)

These movements change the drone's position without changing its orientation.

- **X-axis (Surge):** Forward and backward movement.
- **Y-axis (Sway):** Left and right movement.
- **Z-axis (Heave):** Upward and downward movement (change in altitude).

2. Rotational Movements (Angular Motion)

These movements change the drone's orientation.

- **Roll (X-axis):** Tilting the drone to the left or right.
- **Pitch (Y-axis):** Tilting the drone forward or backward.
- **Yaw (Z-axis):** Rotating the drone clockwise or counter-clockwise about its vertical axis.

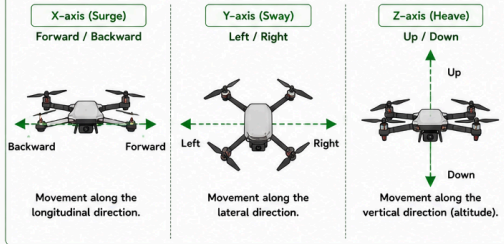
Conclusion

The combination of these six movements allows a drone to perform stable hovering, directional flight, smooth turns, and precise navigation. The flight controller continuously processes sensor data from the IMU to control these six degrees of freedom.

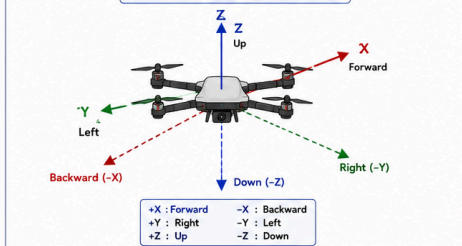
6 DEGREES OF FREEDOM (6DOF) OF A DRONE

6DOF describes the ability of a drone to move and orient itself freely in three-dimensional space. It consists of 3 translational movements (linear motion) and 3 rotational movements (angular motion). These are measured using an Inertial Measurement Unit (IMU) which includes a 3-axis accelerometer and a 3-axis gyroscope.

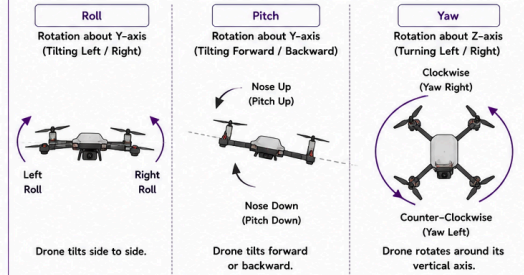
1. TRANSLATIONAL MOVEMENTS (LINEAR MOTION)



3. AXIS REFERENCE (DRONE FRAME)



2. ROTATIONAL MOVEMENTS (ANGULAR MOTION / ATTITUDE)



4. SUMMARY OF 6DOF

Type	Degree of Freedom	Axis	Description
Translational (Linear Motion)	Surge	X-axis	Forward / Backward
	Sway	Y-axis	Left / Right
	Heave	Z-axis	Up / Down (Altitude)
Rotational (Angular Motion)	Roll	Rotation about X-axis	Tilting Left / Right
	Pitch	Rotation about Y-axis	Tilting Forward / Backward
	Yaw	Rotation about Z-axis	Turning Left / Right

★ **NOTE:** The combination of these 6 degrees of freedom allows a drone to hover, move in any direction, tilt, turn and perform stable flight in 3D space.

Q11. Explain roll, pitch, yaw, surge, sway, and heave with examples of each movement in a drone.

A drone moves in three-dimensional space using **6 Degrees of Freedom (6DOF)**, which consists of **three rotational movements** and **three translational movements**. These movements are achieved by varying the speed of individual motors, allowing the drone to change its orientation and position accurately.

A. Rotational Movements (Attitude Control)

Rotational movements change the orientation of the drone about its three axes.

1. Roll (Rotation about the X-axis)

Roll is the tilting movement of the drone to the left or right about its longitudinal (X) axis. It is produced by increasing the speed of the motors on one side of the drone while decreasing the speed on the opposite side.

Example: A drone performs a roll maneuver to counter crosswinds or to begin moving sideways while maintaining its forward direction.

2. Pitch (Rotation about the Y-axis)

Pitch is the forward or backward tilting of the drone about its lateral (Y) axis. To pitch forward, the rear motors rotate faster than the front motors, causing the drone to tilt and move forward.

Example: An agricultural drone pitches forward while flying across a field to spray crops efficiently.

3. Yaw (Rotation about the Z-axis)

Yaw is the rotation of the drone about its vertical (Z) axis. It is achieved by creating a torque difference between the clockwise and counter-clockwise rotating propellers.

Example: A camera drone uses yaw to rotate and capture a 360° panoramic view without changing its position.

B. Translational Movements (Linear Motion)

Translational movements change the position of the drone without necessarily changing its orientation.

1. Surge (Movement along the X-axis)

Surge is the forward or backward movement of the drone. It is usually produced by maintaining a pitch angle that directs part of the thrust horizontally.

Example: A delivery drone surges forward while travelling between two GPS waypoints.

2. Sway (Movement along the Y-axis)

Sway is the left or right movement of the drone. It is achieved by maintaining a roll angle that shifts the drone sideways.

Example: A drone performs a sway movement to avoid an obstacle detected during flight.

3. Heave (Movement along the Z-axis)

Heave is the upward or downward movement of the drone, resulting in a change in altitude. It occurs when the total thrust becomes greater than or less than the drone's weight.

Example: During take-off, a drone performs a heave movement to rise from the ground and reach its desired hovering altitude.

Conclusion

The combination of **roll, pitch, yaw, surge, sway, and heave** enables a drone to fly smoothly in any direction, maintain stability, avoid obstacles, and perform autonomous navigation with precision.

Q12. Explain the complete signal flow in a drone: Pilot Input to Radio Transmitter to Receiver to Flight Controller to ESC to BLDC Motors to Propellers to Drone Movement. Illustrate with a block diagram.

The signal flow in a drone describes how the pilot's commands are transmitted, processed, and converted into physical movement. It involves coordination between the ground control system and the onboard electronic components to ensure stable and controlled flight.

Signal Flow

1. **Pilot Input:**
The process begins when the pilot operates the **Remote Controller (RC)** or **Ground Control Station (GCS)** by moving the joysticks to control the drone's altitude, direction, speed, or orientation.
2. **Radio Transmitter:**
The remote controller converts the pilot's joystick movements into digital control signals and transmits them wirelessly as radio waves to the drone.
3. **Radio Receiver:**
The onboard radio receiver receives these wireless signals and forwards them to the flight controller for processing.
4. **Flight Controller (FC):**
The **Flight Controller (FC)** acts as the brain of the drone. It processes the pilot's commands together with real-time sensor data from the IMU, GPS, and other onboard sensors. Based on this information, it generates **Pulse Width Modulation (PWM)** signals to control the speed of each motor.
5. **Electronic Speed Controller (ESC):**
The **Electronic Speed Controller (ESC)** receives the PWM signals from the flight controller and regulates the electrical power supplied to each Brushless DC (BLDC) motor. This controls the motor's rotational speed (RPM).
6. **BLDC Motors:**
The **Brushless DC (BLDC) motors** convert electrical energy into mechanical rotation. Their speed is continuously adjusted by the ESCs according to the commands from the flight controller.
7. **Propellers:**
The rotating propellers generate lift and thrust by pushing air downward. By varying the speed of individual propellers, the drone can climb, descend, hover, and move in different directions.
8. **Drone Movement:**
The combined thrust produced by the propellers enables the drone to move in three-dimensional space. It performs movements such as **roll, pitch, yaw, surge, sway, and heave**, allowing stable flight and precise maneuvering.

Working Principle

The pilot's commands are transmitted wirelessly to the drone through the radio transmitter and receiver. The flight controller processes these commands along with sensor feedback to maintain stability and generate PWM signals. The ESCs regulate the speed of the BLDC motors, which rotate the propellers to produce the required lift and thrust. By adjusting the speed of each motor independently, the drone performs controlled movements and maintains stable flight under different operating conditions.

Q13. Why is PID control important in drones? Explain the proportional, integral, and derivative terms.

A **PID (Proportional-Integral-Derivative) controller** is one of the most important control algorithms used in drones. It helps maintain stable flight by continuously comparing the drone's desired position or orientation with its actual position and making the necessary corrections.

Since drones are affected by factors such as wind, payload changes, and sudden movements, the PID controller ensures smooth, balanced, and accurate flight.

Importance of PID Control

The PID controller works as a **closed-loop feedback system**. It receives real-time data from sensors such as the **IMU, GPS, and barometer**, calculates the error between the desired and actual state, and adjusts the motor speeds accordingly. Its importance includes:

- **Maintains Flight Stability:** Keeps the drone balanced during take-off, hovering, landing, and maneuvering.
- **Minimizes Error:** Continuously reduces the difference between the desired position and the actual position of the drone.
- **Compensates for Disturbances:** Automatically corrects the drone's motion when affected by wind, external forces, or changes in payload.
- **Provides Smooth Control:** Generates precise PWM signals for the ESCs to regulate motor speed, resulting in stable and smooth flight.
- **Improves Accuracy:** Enables precise altitude control, attitude stabilization, and waypoint navigation during autonomous missions.

Components of PID Controller

1. Proportional (P) Control

The **Proportional** term produces a correction proportional to the current error. A larger error results in a stronger corrective action, allowing the drone to respond quickly to disturbances. However, if the proportional gain is too high, the drone may overshoot its target and oscillate.

2. Integral (I) Control

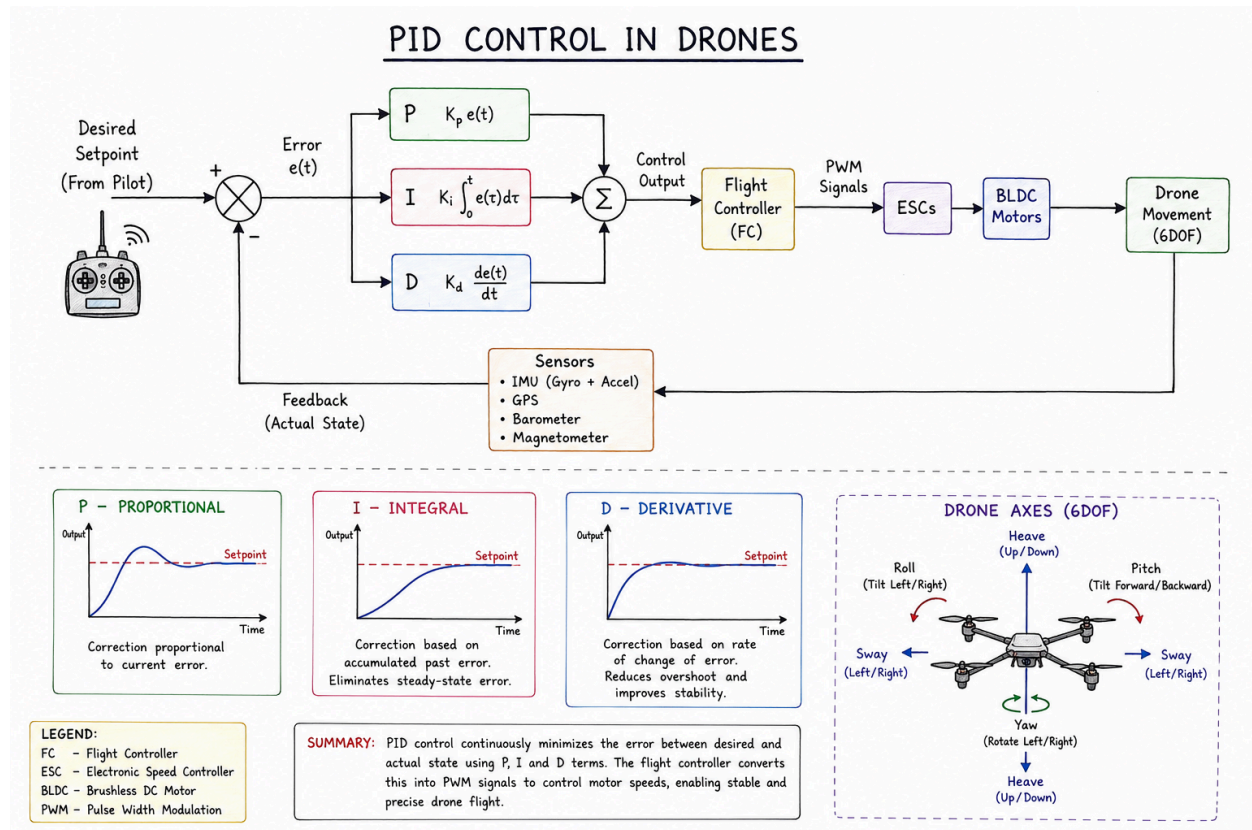
The **Integral** term considers the accumulated error over time. It eliminates steady-state errors that cannot be corrected by proportional control alone, such as a constant drift caused by wind or slight imbalance. This helps the drone maintain its desired position accurately.

3. Derivative (D) Control

The **Derivative** term predicts future error by measuring how quickly the error is changing. It acts as a damping factor that slows the drone as it approaches the desired position, reducing overshoot and minimizing oscillations for smoother flight.

Conclusion

The combined action of the **Proportional, Integral, and Derivative** terms enables a drone to respond quickly, maintain stability, eliminate steady-state errors, and perform smooth, accurate maneuvers. For this reason, PID control is widely used in modern drone flight controllers for both manual and autonomous flight operations.



Q14. Bonus — Research Questions. Answer any TWO of the following:

- Compare DJI, ArduPilot, and PX4 flight controllers.
- Explain obstacle avoidance in drones.
- How do drones return to home automatically?
- Explain drone swarm technology.
- What is autonomous navigation?
- Explain computer vision in drones.
- Discuss future trends in drone technology.

(a). Explain Obstacle Avoidance in Drones.

Obstacle avoidance is a safety feature that enables a drone to detect obstacles in its surroundings and automatically change its flight path to prevent collisions. Also known as **Sense and Avoid (SAA)** or **Detect and Avoid (DAA)**, this technology allows drones to fly safely in both manual and autonomous missions by continuously monitoring the environment and making real-time navigation decisions.

1. Sensing Technologies

Drones use different sensors to detect nearby obstacles depending on the application and operating environment.

- **Ultrasonic Sensors:** Measure the distance to nearby objects by transmitting ultrasonic waves and calculating the time taken for the reflected signal to return. They are suitable for short-range obstacle detection and precision landing.
- **LiDAR (Light Detection and Ranging):** Uses laser pulses to measure distances and generate accurate three-dimensional maps of the surroundings. LiDAR provides high precision and is widely used for autonomous navigation and mapping.
- **Camera and Computer Vision:** Cameras capture images of the environment, while image processing and Artificial Intelligence (AI) algorithms identify and track obstacles in real time.
- **Radar:** Uses radio waves to detect objects over longer distances and performs reliably under different weather conditions. It is commonly used for detecting other aircraft.
- **ADS-B (Automatic Dependent Surveillance-Broadcast):** Receives position, altitude, and velocity information transmitted by nearby aircraft, allowing the drone to maintain safe separation from both manned and unmanned aircraft.

2. Path Planning and Obstacle Avoidance

After detecting an obstacle, the drone determines a safe path using obstacle avoidance algorithms.

- **Reactive Logic:** The drone immediately changes its direction or stops when an obstacle is detected within a predefined safety distance.
- **Potential Field Method:** The destination acts as an attractive force, while obstacles act as repulsive forces. The drone follows the safest path by balancing these virtual forces.
- **Wavefront Planner:** The environment is divided into small grids, and the shortest obstacle-free path is selected before the drone moves.
- **Deconfliction Algorithms:** Advanced systems predict possible collisions with other aircraft and generate safe avoidance maneuvers to maintain minimum separation distances.

3. Hardware Implementation

Modern autonomous drones use a layered control architecture.

- A **Companion Computer** (such as Raspberry Pi or NVIDIA Jetson) processes sensor data, performs image processing, and executes obstacle avoidance algorithms.
- The **Flight Controller** (such as Pixhawk) receives the avoidance commands and controls the motors through the Electronic Speed Controllers (ESCs) to safely change the drone's direction.

4. Challenges and Limitations

Although obstacle avoidance systems have improved significantly, they still face several challenges:

- Limited processing power for complex algorithms.
- Sensor errors caused by rain, fog, reflective surfaces, or poor lighting.
- GPS inaccuracies in urban areas or dense forests.

- Communication delays between the companion computer and the flight controller.
- Difficulty navigating highly cluttered or dynamic environments with multiple moving obstacles.

Conclusion

Obstacle avoidance is an essential feature of modern drones that enhances flight safety and enables autonomous navigation. By combining advanced sensors, intelligent path-planning algorithms, and real-time flight control, drones can detect obstacles, avoid collisions, and complete missions more safely and efficiently.

(b). How do drones return to home (RTH) automatically?

Return-to-Home (RTH), also known as **Return-to-Launch (RTL)**, is an autonomous safety feature that enables a drone to automatically return to its take-off location or a predefined home point without pilot intervention. The home position is recorded using GPS when the drone takes off. RTH is widely used as a safety mechanism during emergencies and autonomous missions.

1. Triggers for Return-to-Home

The Return-to-Home function can be activated under the following conditions:

- **Mission Completion:** After completing a waypoint mission, the drone automatically returns to its home location.
- **Manual Activation:** The pilot can activate RTH using the remote controller or ground control station.
- **Failsafe Conditions:** The flight controller automatically initiates RTH when the battery level becomes critically low or when communication with the remote controller is lost.
- **Emergency Situations:** In certain cases of signal interference or unexpected operating conditions, the drone may also initiate the Return-to-Home procedure.

2. Working of Return-to-Home (RTH)

The Return-to-Home process is controlled by the flight controller using data from the **GPS module**, **Inertial Measurement Unit (IMU)**, and **barometer**.

The sequence is as follows:

1. The drone records the **home point** during take-off.
2. When RTH is triggered, the drone first climbs to a predefined **safe altitude** to avoid obstacles such as trees and buildings.
3. The flight controller uses GPS coordinates to calculate the safest path back to the recorded home location.
4. During the return flight, the flight controller continuously monitors its position and corrects its path using sensor feedback.
5. Once the drone reaches the home point, it performs a controlled landing by gradually reducing motor speed until it safely touches the ground.

3. Challenges and Limitations

Although Return-to-Home is a reliable safety feature, it has certain limitations:

- It depends on accurate GPS signals; weak or interrupted signals may reduce navigation accuracy.
- Basic RTH systems may not detect newly appearing obstacles unless obstacle avoidance sensors are available.
- Strong winds and harsh weather conditions can affect the drone's return path and landing accuracy.

Conclusion

Return-to-Home (RTH) is an essential autonomous feature that improves the safety and reliability of drone operations. By combining GPS navigation, onboard sensors, and intelligent flight control algorithms, the drone can safely return to its launch point with minimal pilot intervention.

Q15. Practical Activity — Choose ONE:

- **Identify and label every component of a real drone or a drone image.**
- **Create a 3D CAD model or hand sketch of a quadcopter.**
- **Use a drone simulator and document take-off, hover, roll, pitch, yaw, and landing with screenshots/video and observations.**

(a) Identify and label every component of a real drone or a drone image.

A drone consists of several mechanical, electrical, and electronic components that work together to achieve stable flight, navigation, communication, and payload operation. Each component performs a specific function, enabling the drone to take off, maintain balance, execute autonomous missions, and land safely. Proper identification of these components is essential for understanding the construction and operation of unmanned aerial vehicles (UAVs).

The major components of a drone and their functions are listed below:

Component	Function
Frame	Forms the structural body of the drone and supports all other components.
Propellers	Generate lift and thrust by rotating at high speed.
BLDC Motors	Drive the propellers to produce the required thrust.
Electronic Speed Controllers (ESCs)	Control the speed of each motor according to commands from the flight controller.
Flight Controller	Acts as the brain of the drone by processing sensor data and controlling flight stability.

LiPo Battery	Supplies electrical power to the drone and all onboard electronics.
GPS Module	Provides location, navigation, and supports autonomous flight modes such as Return-to-Home (RTH).
IMU (Inertial Measurement Unit)	Measures acceleration and angular motion to maintain stability and orientation.
Camera	Captures images and videos for surveillance, mapping, inspection, or photography.
Gimbal	Stabilizes the camera to produce smooth and vibration-free footage.
Landing Gear	Supports the drone during take-off and landing while protecting its components.
Antenna / Receiver	Receives commands from the remote controller and enables wireless communication.

Conclusion

Each component of a drone plays an important role in ensuring safe, stable, and efficient flight. Together, these components allow the drone to perform various applications such as aerial photography, surveying, agriculture, inspection, disaster management, and autonomous navigation.

COMPONENTS OF A REAL DRONE (QUADCOPTER)



- | | |
|--|--|
| 1. Propellers – Generate lift and thrust. | 7. IMU – Measures acceleration and angular velocity. |
| 2. BLDC Motors – Rotate the propellers. | 8. Camera – Captures images and videos. |
| 3. Electronic Speed Controllers (ESCs) – Control motor speed. | 9. Gimbal – Stabilizes the camera. |
| 4. Frame / Body – Provides structural support. | 10. GPS Module – Provides location and navigation. |
| 5. LiPo Battery – Supplies power to the drone. | 11. Antennas / Receiver – Enables communication. |
| 6. Flight Controller – Processes data and controls the drone. | 12. Landing Gear – Supports the drone during landing. |

Higher-Order Thinking Questions

Reflect on and answer the following in your own words: (NO AI Based Answers)

1. Why do quadcopters require four motors instead of one?

A quadcopter requires four motors because a single motor cannot provide the stability and control needed for flight. Each motor can be controlled independently, allowing the drone to balance itself and perform movements such as roll, pitch, and yaw. By increasing or decreasing the speed of specific motors, the flight controller can change the drone's direction while keeping it stable.

The motors are arranged so that two rotate clockwise and the other two rotate counterclockwise. This balances the torque produced by the propellers and prevents the drone

from spinning uncontrollably. If a quadcopter had only one motor, it would generate lift but would not be able to maintain balance or control its orientation during flight.

Therefore, using four motors provides the stability, maneuverability, and precise control required for safe and efficient drone operation.

2. What would happen if one motor stops working during flight?

If one motor stops working during flight, the quadcopter becomes unbalanced because the lift and torque are no longer evenly distributed. The remaining three motors cannot fully compensate for the missing thrust, so the drone will start tilting and may begin spinning uncontrollably.

In most cases, the flight controller will try to stabilize the drone by adjusting the speed of the remaining motors. However, a standard quadcopter usually cannot maintain stable flight with only three working motors. As a result, it is likely to lose altitude and eventually crash if an emergency landing is not possible.

This is why regular maintenance and checking the condition of the motors and propellers before every flight is important for safe drone operation.

3. Why are brushless motors preferred over brushed motors?

Brushless motors are preferred because they are more efficient, reliable, and have a longer lifespan than brushed motors. Since they do not have brushes, there is less friction and wear, which means they require less maintenance and can operate at higher speeds.

For drones, brushless motors also provide smoother and more precise control, allowing the flight controller to make quick adjustments for stable flight. They generate less heat and use battery power more efficiently, which helps increase the drone's flight time.

Although brushless motors are more expensive, their better performance, durability, and efficiency make them the preferred choice for most modern drones.

4. Can a drone fly indoors without GPS? Explain.

Yes, a drone can fly indoors without GPS because GPS signals are usually weak or unavailable inside buildings. Instead of GPS, the drone relies on sensors such as the IMU, optical flow sensor, ultrasonic sensor, or LiDAR to maintain its position and stability. The flight controller uses data from these sensors to keep the drone balanced and navigate safely. However, indoor flight may be less accurate if the drone does not have enough sensors or if the environment has poor lighting.

5. Why are Li-Po batteries used in drones instead of standard batteries?

Li-Po (Lithium Polymer) batteries are preferred because they are lightweight, have a high energy density, and can deliver large amounts of current in a short time. This allows the motors to generate enough thrust for take-off and rapid movements. Compared to standard batteries, Li-Po batteries provide better performance and longer flight time while adding less weight to the drone.

6. What factors reduce a drone's flight time?

Several factors can reduce a drone's flight time, including a low-capacity battery, carrying heavy payloads, strong winds, high flying speed, frequent acceleration, and poor battery condition. Cold weather can also reduce battery efficiency. Since the motors have to work harder under these conditions, the battery drains more quickly.

7. How can drone stability be improved during strong winds?

Drone stability during strong winds can be improved by using a well-tuned flight controller with properly adjusted PID values. GPS, IMU, and other onboard sensors continuously detect changes in the drone's position and orientation, allowing the flight controller to make quick corrections. Reducing speed, maintaining a safe altitude, and avoiding flying in extremely windy conditions also help improve stability.

8. Why is sensor fusion important in drones?

Sensor fusion is important because no single sensor can provide completely accurate information under all conditions. By combining data from multiple sensors such as the GPS, IMU, barometer, and magnetometer, the flight controller obtains a more accurate estimate of the drone's position, altitude, and orientation. This improves flight stability, navigation accuracy, and overall safety.

9. Explain how GPS and IMU work together for stable navigation.

GPS provides the drone's position, speed, and direction over long distances, while the IMU measures acceleration and angular movement in real time. The flight controller combines data from both sensors to accurately determine the drone's location and orientation. The IMU provides fast updates even when GPS signals are weak, while GPS corrects any drift that may develop over time. Together, they help the drone navigate accurately and maintain stable flight.

10. If you were designing a drone for mountain rescue operations, which sensors and components would you choose, and why?

If I were designing a drone for mountain rescue operations, I would choose a GPS module for navigation, an IMU for stable flight, LiDAR and ultrasonic sensors for obstacle detection, and a thermal camera to locate people in low visibility or at night. A high-resolution camera would help rescuers inspect difficult terrain, while a long-range communication module would maintain contact with the ground station. I would also use a high-capacity Li-Po battery, brushless motors, and a strong but lightweight frame so the drone could fly longer and carry emergency supplies if required. These components would make the drone reliable, safe, and effective for search and rescue missions in challenging environments.